

CHULALONGKORN UNIVERSITY

Design and Development of a Virtual Reality Earthquake Response Simulation

for Human-Behavior Study in a Thai Engineering Laboratory Context

ICE Pre-Project Proposal | Course 2143491 | Academic Year 2026

Project administration	Complete before submission
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Committee member	[Insert advisor-proposed committee member]
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Abstract

Earthquakes create fast, uncertain decisions while falling or sliding non-structural objects can obstruct safe action. Conventional drills cannot reproduce these hazards safely or with consistent experimental control. This project proposes CEVR, a Unity and OpenXR virtual-reality simulation that places a participant in a fictional engineering teaching laboratory, assigns ordinary tasks, introduces a delayed earthquake, records protective and evacuation behavior, and provides a controlled post-event assembly objective. The technical design separates a direct-instruction Training mode from a neutral-prompt Research mode. Physical objects receive acceleration from either a deterministic tutorial preview or a validated recorded ground-motion profile; the participant camera is never shaken. The proposed study uses a 2 x 2 between-participant design to examine two calibrated shaking-intensity bands and two floor-response conditions. Primary outcomes are protective-action latency, cover choice, unsafe exit, evacuation time, collisions, and completion. Secondary measures address presence, perceived realism, and simulator sickness. The system logs anonymous semantic events and head/hand poses while enforcing emergency-stop and data-minimization rules. The expected contribution is a reproducible, culturally contextualized prototype and research workflow for examining earthquake response in a Thai university setting without exposing participants to a real hazard. The proposal explicitly excludes structural-damage prediction and requires ethics approval, device validation, power analysis, and pilot testing before human data collection.

Keywords: immersive virtual reality; earthquake preparedness; human behavior; emergency evacuation; serious games; Unity; OpenXR; non-structural hazards

Document Control and Course Compliance

Requirement	Current status	Required action
Topic confirmation by 28 Aug 2026, 23:59	Draft topic and implementation package prepared	Advisor must approve or revise the Moodle topic
Group size 2-5 ICE students	Administrative data not supplied	Insert all members and IDs
Advisor supervision limit: three groups	Not confirmed	Confirm capacity with advisor
Proposal report by 20 Nov 2026, 23:59	Complete draft supplied	Revise with advisor and submit through Moodle
Proposal conference 7-8 Dec 2026	Presentation not yet prepared	Create presentation after proposal approval
Turnitin similarity below 25%	Not yet checked	Run Turnitin and attach report as final page

Course rule: The orientation states that hard deadlines are non-negotiable except special cases and that late submission deducts 10 marks each time [1].

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1. Introduction

1.1 Background

During an earthquake, occupants may have only seconds to choose between protective action, movement, and evacuation. Recommended guidance emphasizes dropping, taking cover beneath sturdy furniture, and holding on until shaking stops [2]. In actual buildings, however, furniture quality, room density, overhead equipment, and available cover differ by local context. A teaching laboratory may contain tall cabinets, suspended lights, electronic equipment, benches, and narrow circulation routes. These non-structural features affect both perceived risk and feasible action.

Virtual reality can expose participants to a repeatable emergency scenario while preserving experimental control and avoiding physical earthquake motion. Prior work has used immersive serious games for earthquake preparedness and reported improvements in knowledge and self-efficacy [4]. Other work has used immersive virtual reality and verbal protocol analysis to investigate decisions during shaking and post-earthquake evacuation [5]. These studies demonstrate potential, but behavioral validity, simulator sickness, experimental priming, and differences between virtual and real movement remain important limitations [7, 8, 9].

The proposed project focuses on a Thai university engineering context. It does not reproduce a specific campus building. Instead, it examines how a compact laboratory layout, shared cover, furniture strength, overhead hazards, and floor response can be represented in a controlled prototype. This localization differentiates the project from previous hospital and generic-building simulations without claiming that a visual difference alone constitutes scientific novelty.

1.2 Problem Statement

There is no readily available, research-ready Unity stage tailored to the project's intended Thai engineering-laboratory context that simultaneously provides deterministic ground-motion playback, neutral and instructional study modes, physical-object hazards, cover detection, safe post-event evacuation logic, anonymous event logging, reproducible GitHub collaboration, and explicit VR comfort controls. A simple script that applies random force to every rigidbody is insufficient because it is difficult to reproduce, does not represent recorded acceleration, can bias outcomes through artificial cues, and may incorrectly imply structural accuracy.

1.3 Aim

The project aims to design, implement, and evaluate an immersive VR earthquake-response simulation that can support both a guided tutorial and a future approved behavioral experiment in a fictional Thai engineering laboratory.

1.4 Objectives

1. Develop a complete Unity stage with ordinary laboratory tasks, delayed earthquake onset, physical hazards, protective cover, health, evacuation, and debrief states.
2. Implement deterministic tutorial motion and a validated importer for recorded three-axis acceleration data.
3. Avoid camera motion and forced locomotion while providing environmental audio, lighting, and object-response cues.
4. Record anonymous behavioral events and head/hand poses in a machine-readable format.
5. Separate direct Training prompts from neutral Research prompts to reduce experimental priming.
6. Evaluate functional correctness, usability, comfort, performance, and research feasibility through staged verification and pilot testing.
7. Maintain an organized GitHub repository that supports reproducible team collaboration and review.

1.5 Research Questions and Hypotheses

ID	Question or hypothesis
RQ1	How does calibrated shaking intensity affect latency and selection of protective action?
RQ2	How does a lower-floor versus upper-floor response profile affect perceived intensity and behavior?
RQ3	How do furniture location, cover capacity, and falling-object hazards influence cover and exit decisions?
RQ4	Can the simulation operate with acceptable usability, frame pacing, presence, and simulator-sickness scores?
H1	The higher-intensity condition will produce shorter protective-action latency and higher perceived urgency.
H2	The upper-floor response condition will increase perceived motion intensity relative to the lower-floor condition.
H3	Participants with immediately accessible sturdy cover will be more likely to use cover than to attempt an early exit.

1.6 Scope and Exclusions

- One fictional laboratory and adjacent outdoor assembly area form the MVP.

- The simulation models non-structural object response and participant decisions, not structural failure.
- Local shaking is represented by acceleration time history and calibrated floor response, not earthquake magnitude alone.
- Eye tracking, biometric sensing, crowds, full glass fracture, and multi-room building evacuation are excluded from the MVP.
- The first evaluation is a controlled pilot; claims about real-world safety effectiveness are outside scope.

2. Literature and Technical Review

2.1 Earthquake Motion and Non-Structural Hazards

Peak ground acceleration is a useful descriptor of local shaking but does not independently describe frequency content, duration, direction, site effects, or building response. USGS ShakeMap combines observations and models to describe ground motion and shaking intensity for response and planning [3]. For this reason, CEVR stores three-axis acceleration time histories in metres per second squared and records profile identity. A reduced single-degree-of-freedom floor-response component is available, but it must be calibrated before research use.

The simulation distinguishes sliding, rocking/toppling, falling, and suspended-object motion through Rigidbody mass, collider geometry, center of mass, joints, and inertial acceleration. No random torque is added to make objects fail. Hazard order is serialized so repeated trials request the same release sequence.

2.2 Earthquake Protective Action

FEMA guidance recommends Drop, Cover, and Hold On for most indoor earthquake situations and emphasizes remaining under sturdy furniture until shaking stops [2]. CEVR teaches that sequence only in Training mode. Research mode does not display the instruction at onset because doing so would convert the dependent behavior into a prompted response. The project also avoids treating an early outdoor run as automatic success; the attempt is logged separately and post-shaking evacuation is evaluated as a later phase.

2.3 Immersive VR for Emergency Research

Immersive VR permits controlled exposure to complex hazards that are difficult or unethical to reproduce physically. Feng and colleagues developed an earthquake serious game and found increased preparedness knowledge and self-efficacy after training [4]. Their decision-making study showed that

think-aloud protocols can expose reasoning during virtual earthquake and post-earthquake evacuation [5]. Kinader and colleagues demonstrated how social influence can be examined in real and virtual evacuation contexts [7].

Nevertheless, virtual behavior is not automatically equivalent to real behavior. Comparative evacuation research reports that head-mounted VR can reproduce some outcomes, such as pre-evacuation time and exit choice, while movement and orientation differences remain [8]. The proposed work therefore describes observed behavior inside the simulation and avoids claiming direct real-world equivalence without validation.

2.4 Cybersickness, Presence, and Validity

Simulator sickness can include nausea, oculomotor discomfort, and disorientation. The Simulator Sickness Questionnaire provides a recognized symptom framework [9]. CEVR reduces visual-vestibular conflict by keeping the camera controlled only by the tracked head, shaking the environment and objects rather than the participant viewpoint, using a world-space HUD, limiting scenario duration, and providing immediate stop controls. Presence and realism should be measured separately from sickness because a visually convincing scene can still produce discomfort or behavior that differs from the real world.

2.5 Thai Context and Research Gap

A 2026 study of Thai dental students following a major regional earthquake reported preparedness and stress outcomes and found higher-floor residence associated with greater post-traumatic stress [13]. This does not prove a causal floor effect in VR, but it supports examining floor context as a relevant participant variable. Chulalongkorn University has also provided earthquake-awareness guidance following the 2025 regional event [14]. The proposed contribution is not merely Thai decoration; it is a reproducible framework for testing contextual room constraints, calibrated floor response, and behavior metrics with clear limits on inference.

3. Proposed System

3.1 Technology Stack

Component	Selected technology	Rationale
Engine	Unity 6000.3.20f1	Pinned project version and physics/editor ecosystem
XR runtime	OpenXR Plugin 1.18.0	Headset-independent runtime with project validation

Component	Selected technology	Rationale
Interaction	XR Interaction Toolkit 3.1.3	Component-based grab, controller, and simulator support [10, 11]
Input	Unity Input System 1.14.2	Desktop debug and XR action support
Testing	Unity Test Framework 1.6.0	EditMode rule and data-model tests
Collaboration	GitHub plus Git LFS	Text scene review, issue ownership, CI, and binary asset handling [12]
Data	UTF-8 JSONL	Appendable, parseable event records with unique session files

3.2 Gameplay Sequence

8. Orientation: the participant observes the laboratory and learns locomotion and stop controls.
9. Normal activity: the participant places a circuit module and safety canister while an unseen 30-second minimum timer runs.
10. Earthquake: deterministic motion starts; lights flicker, rumble plays, physical objects respond, and hazards release in a fixed sequence.
11. Protective behavior: cover-zone entry, unsafe exit, collisions, and damage are recorded.
12. Post-event evacuation: after shaking stops, the assembly point becomes eligible for success.
13. Outcome and debrief: the system records success, health depletion, timeout, or participant/facilitator stop.

3.3 Software Modules

Module	Responsibility	Research-control feature
GameFlowController	Phase state machine and outcome	Single source of phase truth
TutorialScenarioConfig	Timing, mode, prompts, rules	Versionable scenario asset
GroundMotionPlayer	Recorded or preview acceleration	Profile identity and deterministic seed
InertialRigidbody	Apply acceleration to objects	No camera transform and no random torque
HazardDirector	Release five hazards	Serialized order and time
Cover/Assembly zones	Protective and success triggers	Semantic entry events
SessionLogger	Unique JSONL sessions	Participant-code sanitation
PoseTelemetrySampler	Head and hand position at 10 Hz	Fixed, documented sample request

Module	Responsibility	Research-control feature
Stage Builder	Generate scene and wiring	Clean-clone reproducibility

3.4 Scene Content

The generated room includes a floor, ceiling, walls, windows, concrete columns, laboratory benches, a sturdy cover table, a marked exit, an outdoor walkway, and an assembly point. Dynamic content includes two task objects, two suspended lamps, four overhead hazards, and one unsecured cabinet. The graybox appearance is intentional: interaction geometry and experimental logic are verified before licensed campus-inspired art is commissioned.

3.5 Motion Model

Recorded profiles contain regularly sampled three-axis acceleration. The importer accepts `g` or SI units, rejects non-finite values, requires increasing time, checks sampling regularity, and stores SI acceleration. Linear interpolation supplies values between samples. In research mode, startup fails if no recorded profile is assigned. A floor-response asset may transform ground acceleration through a reduced-order damped oscillator; its calibration note and profile ID are logged.

Physics limitation: The model controls non-structural object response for an interactive study. It is not finite-element structural analysis and must not be used to certify a building or predict damage.

3.6 Logging and Privacy

Each run creates a new UTF-8 JSONL file under the application persistent-data directory with `FileMode.CreateNew`. Events include UTC timestamp, relative session time, sanitized participant code, scenario ID, build version, event type, and escaped JSON payload. The configuration event records whether preview motion was used, motion-profile ID, floor-response ID, and duration. Raw logs are excluded from Git and must be transferred only under the approved data-management plan.

4. Proposed Research Method

4.1 Study Design

The proposed main study uses a 2 x 2 between-participant design to avoid learning and habituation from repeated earthquakes. Factor A is local shaking intensity; Factor B is floor response. The provisional

target is 64 participants, 16 per condition. This number is a planning value, not a completed power calculation. The final sample must be set through an advisor-approved power analysis using pilot variance and the selected primary outcome.

Condition	Intensity request	Floor-response context	Participants
A	Lower band, approximately 0.08 g peak component	Lower-floor profile	16 provisional
B	Lower band, approximately 0.08 g peak component	Upper-floor calibrated profile	16 provisional
C	Higher band, approximately 0.16 g peak component	Lower-floor profile	16 provisional
D	Higher band, approximately 0.16 g peak component	Upper-floor calibrated profile	16 provisional

The headset does not physically accelerate the participant. The intensity values control virtual-object acceleration, visual motion cues in the environment, lighting, and audio. Final values must pass pilot comfort and realism checks and must be described as local acceleration conditions, not earthquake magnitude.

4.2 Participants

The planned population is adult university students aged 18 years or older. Recruitment should avoid direct academic coercion and must state that participation or withdrawal has no effect on grades. Exclusion and postponement criteria should include current severe motion sickness, acute vestibular symptoms, uncorrected visual difficulty incompatible with the headset, a medical restriction against VR, or inability to understand the consent and stop procedures. Any health screening must be approved and minimized.

4.3 Variables and Measures

Type	Measure	Operational definition
Primary	Protective-action latency	Seconds from earthquake onset to first crouch or valid cover entry
Primary	Cover behavior	Cover entered; cover duration; first selected protection zone
Primary	Unsafe exit	Assembly-zone entry before post-quake eligibility
Primary	Evacuation time	Seconds from shaking end to valid assembly entry

Type	Measure	Operational definition
Secondary	Collision/damage	Hazard hits, protected state, applied damage, remaining health
Secondary	Task state	Number and identity of tasks complete at onset
Secondary	Simulator sickness	Pre/post symptom questionnaire approved by the protocol
Secondary	Presence and realism	Post-test Likert scales and short comments
Covariate	Prior experience	Previous VR use, earthquake experience, and safety training

4.4 Procedure

14. Obtain written consent, assign a participant code, and complete approved screening and baseline symptoms.
15. Explain guardian boundaries, headset fit, locomotion, and the participant/facilitator stop methods.
16. Run a neutral interaction practice scene that is separate from the earthquake outcome scene.
17. Randomly assign one of four conditions using a concealed schedule, stratified if necessary by prior VR experience.
18. Start the ordinary laboratory activity. Do not announce the exact earthquake onset time.
19. Observe without coaching during the Research-mode event unless safety requires intervention.
20. Stop immediately for participant request or safety concern.
21. After the event, remove the headset safely, assess symptoms, administer questionnaires, and conduct a short debrief.
22. Transfer the anonymous log according to the data plan and record technical anomalies separately.

4.5 Analysis Plan

The team will first publish descriptive distributions, missingness, technical exclusions, and adverse events. Protective-action and unsafe-exit proportions can be evaluated with logistic regression using intensity, floor condition, and their interaction. Continuous latency and evacuation time can be examined with two-way linear models if assumptions are acceptable, or robust/non-parametric alternatives otherwise. Time-to-action analysis may use survival methods when participants never perform an action. Presence and sickness outcomes will be reported with confidence intervals. The pilot is intended to estimate feasibility and variability; it will not be presented as a definitive effectiveness trial.

4.6 Ethics and Safety

The protocol must receive institutional approval before recruitment. Consent must explain the simulated earthquake, falling virtual objects, intense sound, possible discomfort, collected pose data,

voluntary withdrawal, and data retention. A facilitator remains present, and a clear physical area is maintained. The participant can stop without giving a reason. Training data must not be combined with neutral Research-mode data because direct instruction changes behavior.

5. Verification and Evaluation

5.1 Verification Strategy

Level	Method	Acceptance criterion
Repository	Python static verifier and CI	Required files, package pins, JSON, metadata, and policies pass
Unit	Unity EditMode tests	All rule, health, and motion-profile tests pass
Scene	Editor validator	Required controller, player, task, hazard, zone, camera, and listener counts pass
Desktop	Thirteen functional cases	Every state path and log result matches expected behavior
XR simulator	Grab, locomotion, body triggers	Tasks, cover, and assembly work without a headset
Headset	OpenXR validation and device build	Tracking, input, frame pacing, stop, and boundary pass
Pilot	Internal and approved participant pilot	No blocking comfort, data, usability, or validity issue

5.2 Current Implementation Status

Item	Status	Evidence or limitation
Repository structure	Implemented	Runtime, Editor, Tests, docs, CI, and LFS rules included
Gameplay state machine	Implemented	Orientation through Debrief, including stop paths
Motion, hazards, cover, health	Implemented	Deterministic code and static policy check
Anonymous logging	Implemented	Unique JSONL writer and pose sampler
One-click stage builder	Implemented	Editor menu and structural validator included
Static verification	Passed	Available authoring-environment check
Unity compilation	Not yet executed	Unity Editor is unavailable in the authoring environment
OpenXR/headset runtime	Not yet executed	Requires team hardware and selected target platform

Item	Status	Evidence or limitation
Human-participant study	Not started	Requires advisor, ethics, power analysis, and pilot approval

5.3 Performance and Comfort Criteria

- No sustained frame-time violation at the headset refresh target.
- No earthquake-driven camera translation, rotation, or artificial head roll.
- One active XR Origin and one active AudioListener.
- No hazard starts at the initial head position.
- Participant and facilitator emergency-stop controls work in the device build.
- Post-test symptom scores and adverse events remain within advisor- and ethics-approved continuation criteria.

6. Project Management

6.1 Work Plan

Period	Milestone	Deliverable / exit criterion
24-28 Aug 2026	Topic confirmation	Advisor approval in Moodle; group and committee fields completed
29 Aug-13 Sep	Unity foundation	Clean clone compiles; packages and desktop stage pass
14-27 Sep	XR interaction	One XR Origin, grabs, body triggers, stop input, simulator pass
28 Sep-11 Oct	Recorded motion	Importer, selected dataset, units/axes check, floor profiles
12-25 Oct	Instrumentation	Event schema, data dictionary, parser, research prompts
26 Oct-8 Nov	Testing and pilot	Desktop suite, device performance, internal comfort pilot
9-15 Nov	Proposal revision	Advisor comments resolved; references and method frozen
16-19 Nov	Submission QA	Turnitin below 25 percent; final PDF and attachment checked
20 Nov, 23:59	Hard deadline	Proposal submitted on Moodle
21 Nov-6 Dec	Oral preparation	Demo build, presentation, rehearsal, backup video
7-8 Dec	Proposal conference	Presentation delivered in assigned slot

6.2 Proposed Team Roles

Role	Responsibilities	Assigned member
Project lead / research	Advisor coordination, protocol, measures, report, schedule	[Insert name]
Unity systems	State machine, motion, physics, tests, logging	[Insert name]
XR and UX	OpenXR, interaction, comfort, headset builds	[Insert name]
Environment and assets	Lab design, optimization, licensed art, audio	[Insert name]
Data and evaluation	Log parser, statistical plan, QA, reproducibility	[Insert name]

6.3 Resources and Budget

Resource	Requirement	Cost status
VR headset and controllers	OpenXR-compatible device with stable tracking	Confirm existing university/team hardware before purchase
Development workstation	Unity-capable GPU, USB/network deployment	Team or laboratory resource
Unity, OpenXR, XRI	Specified package versions	No license cost expected for student prototype; verify terms
3D/audio assets	Original, CC-compatible, or purchased license	Use graybox first; record every license
Participant support	If approved by ethics/protocol	To be determined with advisor
Data storage	Encrypted institutional location	Use approved university resource

6.4 Risk Register

Risk	Likelihood	Impact	Mitigation / trigger
Cybersickness or panic	Medium	High	No camera shake, short exposure, screening, stop control, facilitator
Headset performance failure	Medium	High	Graybox first, profiler capture, device gate before pilot
Uncalibrated physics misrepresented	Medium	High	Use time history, record units, label prototype limits, advisor review

Risk	Likelihood	Impact	Mitigation / trigger
Research prompt biases action	Medium	High	Separate Training and Research modes; freeze wording
Participant data reaches GitHub	Low	Critical	Gitignore, code-only PRs, access control, incident plan
Unity scene merge conflict	Medium	Medium	Scene owner, Force Text, issue lock, UnityYAMLMerge
Advisor/topic approval delayed	Medium	High	Confirm before 28 Aug; prepare alternate scope
Turnitin similarity exceeds 25%	Low	High	Original writing, early check, attach final report
Sample recruitment insufficient	Medium	Medium	Pilot-first claims, power analysis, extend recruitment window

7. Expected Results and Contributions

The project is expected to produce a functioning VR tutorial stage, a reproducible research-mode configuration, validated motion-import workflow, behavioral event schema, and a documented safety/testing process. The behavioral study is expected to reveal measurable variation in protective-action latency, cover use, unsafe exit, and perceived intensity across conditions. Because those outcomes have not yet been collected, this proposal does not predict statistical significance or claim effectiveness.

- Technical contribution: organized Unity/OpenXR implementation with deterministic physics requests and automated checks.
- Method contribution: separation of training instruction from neutral behavioral observation.
- Context contribution: a fictional Thai engineering-laboratory scenario centered on furniture and cover constraints.
- Reproducibility contribution: pinned dependencies, GitHub workflow, scene generator, tests, and profile-aware logs.
- Educational contribution: a platform that can later be evaluated as an earthquake preparedness training tool.

8. Limitations

Behavior in VR may differ from action during a real earthquake because participants know they are safe, locomotion is mediated, virtual collisions do not cause injury, and the study cannot recreate social

pressure, building noise, dust, power failure, or aftershocks completely. The reduced floor model is not structural analysis. A student sample limits generalizability. Between-participant assignment reduces carryover but requires a larger sample. Presence can increase engagement while also increasing discomfort. Finally, the implemented graybox stage must be compiled and tested in Unity and on the selected headset before it can be considered operational.

9. Conclusion

CEVR proposes a careful middle ground between an entertainment game and an engineering analysis tool. It uses a controlled virtual laboratory to examine observable decisions, applies motion data to objects without shaking the participant viewpoint, records reproducible events, and maintains explicit boundaries around safety and inference. The repository already provides the core architecture and a one-click stage generator; the remaining work is machine-side Unity/XR validation, calibrated condition preparation, advisor and ethics approval, and pilot-led refinement. If these gates pass, the project can provide a credible ICE pre-project foundation and a practical platform for later senior-project research.

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Appendix A. Functional Requirements

ID	Requirement	Acceptance evidence
FR-01	Begin with orientation and ordinary activity before the event	Observed phase sequence and event log
FR-02	Do not start the event before 30 seconds	EditMode timing test and desktop case F02
FR-03	Support two validated ordinary tasks	Correct item IDs complete each goal
FR-04	Apply three-axis motion to dynamic objects without camera motion	Code policy scan and headset observation
FR-05	Release four overhead hazards and one cabinet deterministically	Scene validator and repeated run
FR-06	Reduce damage while in valid cover	Health test and functional case F05
FR-07	Reject assembly success during shaking	Functional case F06
FR-08	Accept assembly after shaking	Functional case F07
FR-09	Provide participant/facilitator stop	Functional case F10 and device control
FR-10	Create unique anonymous session logs	Two-run filename and JSON parse check
FR-11	Reject tutorial preview in Research mode	Functional case F12
FR-12	Generate and validate the stage from a clean clone	Independent reproduction by two team members

Appendix B. Event Data Dictionary

Event	Trigger	Key payload fields
session_started	Log file opens	mode, platform
scenario_configured	Run validates	motionProfile, floorResponse, duration, previewMotion
phase_changed	State transition	phase
task_completed	Correct goal receives item	taskId
earthquake_onset	Motion begins	session time
cover_enter / cover_exit	Body enters or exits cover	zoneld

Event	Trigger	Key payload fields
player_damaged	Hazard hit accepted	damage, remaining, source, protected
unsafe_exit_attempt	Assembly entered too early	zoneld, phase
assembly_enter	Assembly trigger entered	zoneld, accepted
tutorial_success	Valid outcome	healthRemaining
tutorial_failure	Failure completes	reason
trial_aborted	Stop requested	reason
pose_sample	10 Hz telemetry request	head, left hand, right hand XYZ
session_ended	Writer closes	session time

Appendix C. Facilitator Safety Checklist

- Confirm approved protocol version, build hash, device, and assigned condition.
- Clear the physical play area and activate guardian or boundary protection.
- Disinfect and fit the headset; align virtual and real floors.
- Explain voluntary participation and both emergency-stop methods.
- Remain within sight and reach without coaching behavior.
- Stop immediately for request, imbalance, nausea, distress, or technical failure.
- Seat the participant safely before headset removal if symptoms occur.
- Complete post-test symptoms and debrief before the participant leaves.
- Transfer anonymous data and record any deviation or adverse event.

Appendix D. Draft Participant Information Elements

The final participant information sheet must be approved by the responsible institutional process. It should state the study purpose in non-deceptive language, describe the VR earthquake and virtual falling objects, explain possible dizziness or emotional discomfort, identify collected pose and event data, state duration, explain voluntary withdrawal, provide contacts, and describe storage and deletion. If the exact onset is withheld to reduce anticipation bias, the consent and debrief must explain the authorized incomplete disclosure and why it is necessary.

Appendix E. GitHub Definition of Done

- Issue has priority, acceptance criteria, owner, and reviewer.
- Branch contains one coherent scope and no participant data.
- Static verifier, Unity compilation, and relevant tests pass.
- Scene and metadata changes are reviewed for GUID integrity.
- Binary assets use Git LFS and have a documented source and license.
- Headset-impacting changes include device evidence.
- Documentation and changelog reflect externally visible behavior.

Appendix F. Submission Checklist

- Replace all bracketed team, advisor, committee, and hardware fields.
- Confirm the final topic in Moodle by 28 August 2026 at 23:59.
- Obtain advisor review of scope, research questions, design, and claims.
- Update implementation status after Unity and headset gates.
- Run spelling, reference, figure/table, and page-number checks.
- Run Turnitin early enough to revise; similarity must be below 25 percent.
- Export the final PDF and verify every page visually.
- Attach the final Turnitin report as the last page.
- Submit to Moodle by 20 November 2026 at 23:59 and retain proof.

Appendix G. Turnitin Originality Report

Required final attachment: Insert the official Turnitin originality report here after the final manuscript is checked. The ICE orientation requires similarity below 25 percent and requires the Turnitin report to be attached as the last page [1].

This placeholder page is intentionally last. Replace it with the exported Turnitin report before the official proposal submission.